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IBM Quantum Platform

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Operations

q[0] H Z

q[1]

q[2]

c3

Probabilities

Q-sphere

000 001 010 011 100 101 110 111

Computational basis states

Probability (% of 1024 shots)

0 20 40 60 80 100

000 001

Q-sphere

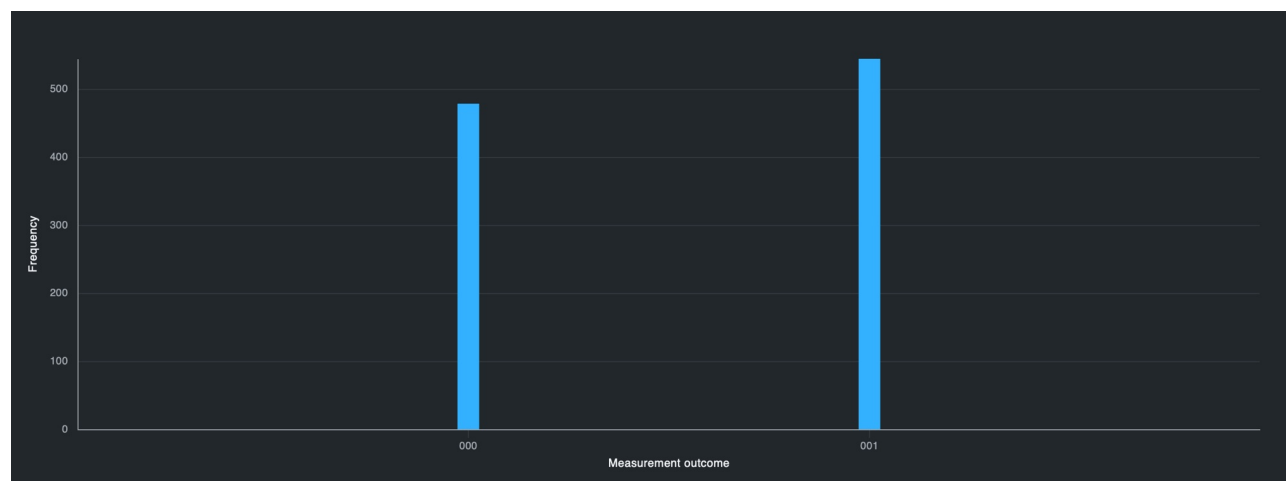
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State ☒ Phase angle ☐

English

Q1, Q2

Q3



```
1 import numpy as np Q4

[ ] 1 v1=np.array([[np.sqrt(3)], [1]])
    2 v2=np.array([[1], [1]])
    3 inner_result=np.vdot(v1,v2)
    4 print(inner_result) Q5

2.732050807568877

[ ] 1 v1=np.array([[1j], [1]])
    2 v2=np.array([[1j], [1]])
    3 inner_result=np.vdot(v1,v2)
    4 print(inner_result) Q6

0j

[ ] 1 v1=np.array([[1+ 2j], [1]])
    2 v2=np.array([[1-1j], [1]])
    3 inner_result=np.vdot(v1,v2)
    4 print(inner_result) Q7

-3j
```

$$5. |\vec{v}_1\rangle = \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} \quad |\vec{v}_2\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \langle \vec{v}_1 | = (\sqrt{3} \quad 1)$$

$$\langle \vec{v}_1 | \cdot |\vec{v}_2\rangle = (\sqrt{3} \quad 1) \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \sqrt{3} + 1 = 1.732 + 1 = 2.732$$

$$6. |\vec{v}_1\rangle = \begin{pmatrix} i \\ 1 \end{pmatrix} \quad |\vec{v}_2\rangle = \begin{pmatrix} -i \\ 1 \end{pmatrix} \quad \langle \vec{v}_1 | = (-i \quad 1)$$

$$\langle \vec{v}_1 | \cdot |\vec{v}_2\rangle = (-i \quad 1) \cdot \begin{pmatrix} -i \\ 1 \end{pmatrix} = -1 + 1 = 0$$

$$7. |\vec{v}_1\rangle = \begin{pmatrix} 1+2i \\ 1 \end{pmatrix} \quad |\vec{v}_2\rangle = \begin{pmatrix} 1-i \\ 1 \end{pmatrix} \quad \langle \vec{v}_1 | = (1-2i \quad 1)$$

$$\langle \vec{v}_1 | \cdot |\vec{v}_2\rangle = (1-2i \quad 1) \cdot \begin{pmatrix} 1-i \\ 1 \end{pmatrix} = 1-i-2i-2+1 = -3i$$